

Enabling Hardware in the Loop with BaSyx and Simulink

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1 Project Overview

1.1 Summary

Simulink could offer a viable simulation environment for execution of Hardware in the Loop using BaSyx to manage Digital Twins and interface with the hardware on the other end.

However, currently, there is no simple alternative of integrating a *Simulink* model and a *BaSyx Asset Administration Shell*, whereby a interface is only realized by use of additional tools and generic closed-source APIs, in expense of simplicity and testability.

A possible solution consists of a custom *Simulink* library capable of managing connections with *BaSyx* Servers and enabling the real-time fetching and update of AAS properties directly within *Simulink*.

This project proposes the development of the afore-described library and the showcase of its usage with a PoC, composed fundamentally by a Simulink Model making use of this custom library and its respective AAS. Test case scenarios will be planned according to the chosen application project.

See 2.3 *Description of Key Terms in Appendix*.

1.2 Problem Definition

- Simulink is highly used to model engineering systems, whose models could be used as a Hardware in the Loop virtual counterpoint for a real system interfacing with a BaSyx Server
- Currently there is no simplified method of realizing the communication between a Simulink Model and a BaSyx AAS Server.

1.3 Proposed Solution

- A Simulink custom library to handle requests to/from a BaSyx server
- A proof-of-concept application project to showcase the usage of the library

For a detailed version of the scope, refer to 2.2 *Detailed Scope*

1.4 Motivation

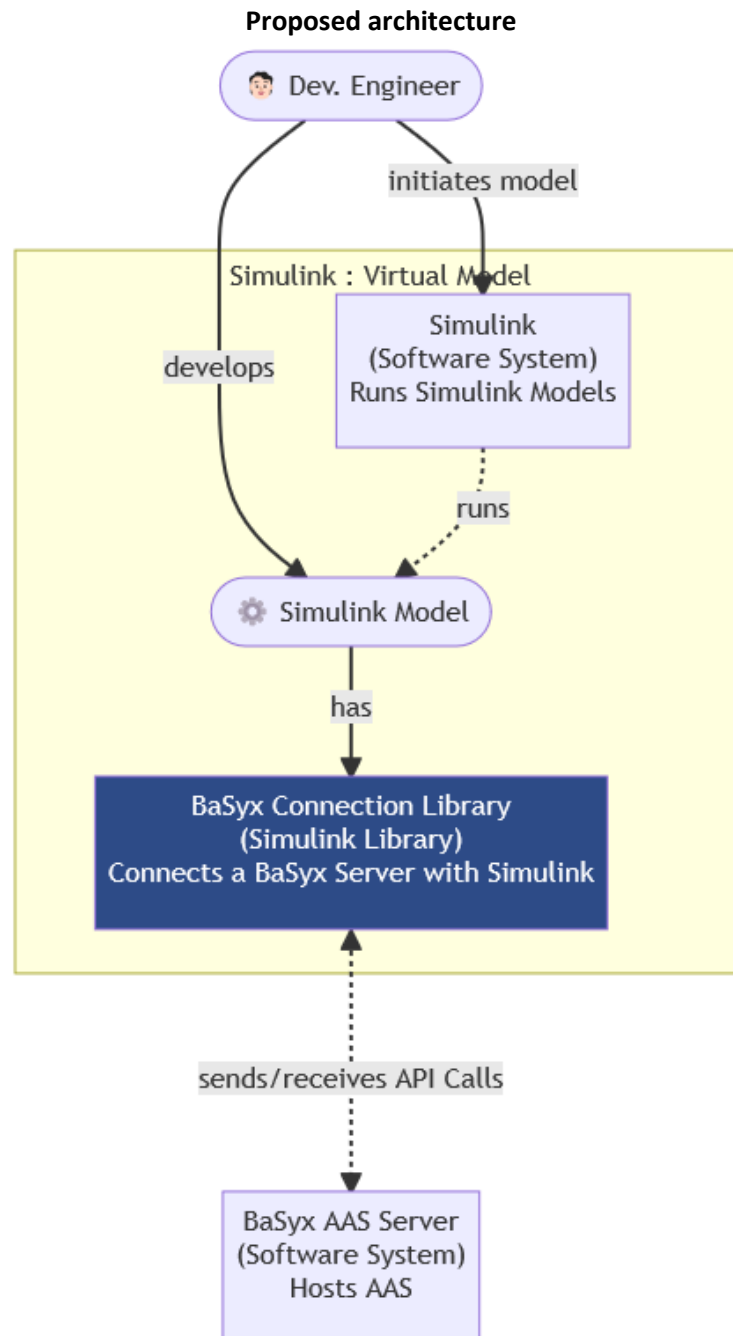
With the successful implementation of this project is expected:

- more incentive to embed digital twin functionalities to new and existing models
- rapid testing of new operational parameters without compromising the real system
- forecasting of relevant metrics based on simulation results
- more agility in structuring integrated Hardware in the Loop solutions
- facilitated real-world data acquisition in virtual models in early phases of development

2 Appendix

2.1 Detailed Scope

- a. A REST based Simulink library using S-Function that can handle requests to/from a set-up BaSyx Server and handle the on-execution and/or real-time update of the model's properties



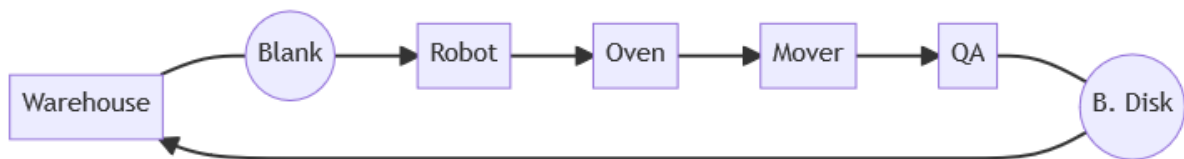
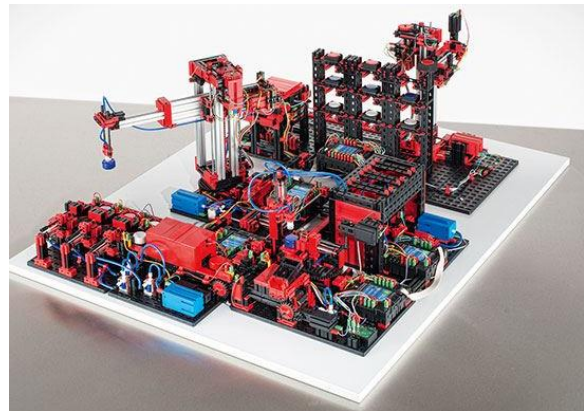
- b. A PoC application project, illustrating the interaction between the Simulink model and a BaSyx AAS.

Refer to 2.2 Proposed application project: Fischertechnik Demonstrator + BaSyx

2.2 Proposed application project: *Fischertechnik* Demonstrator + BaSyx

A physical prototype consisting of the off-the-shelf training and simulation model *Fischertechnik Industrie 4.0* was used to showcase the application of BaSyx in the industry

The learning model simulates the production of a disk brake from its pick-up in the warehouse to its processing, in the oven, assessing of its characteristics in the QA, and finally being stored in the warehouse.

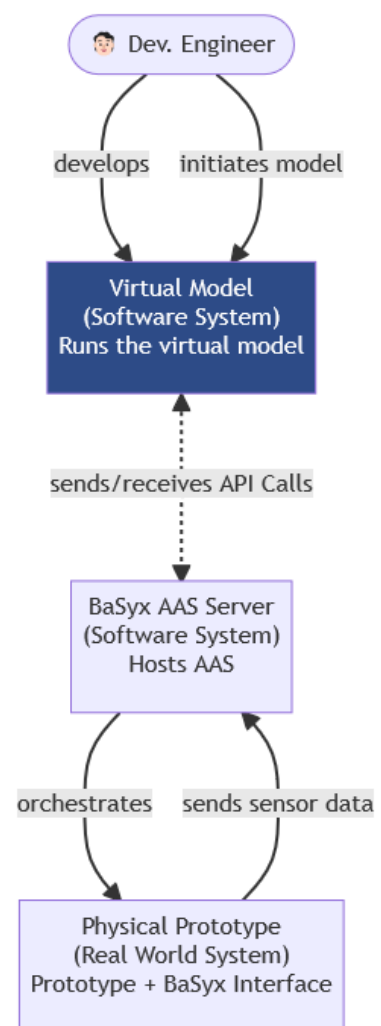


- **Current implementation State**

- The modelling of the presented topology as Asset Administration Shells was implemented by the Fraunhofer IESE team using BaSyx as framework
- The interface between the models and the physical prototype is currently in progress

- **Proposal**

- A virtual model in Simulink of one of the devices that interacts with its digital twin following the diagram on the right.



2.3 Description of Key Terms

Eclipse BaSyx

Eclipse BaSyx is a middleware that supports the implementation of Industrie 4.0 production environments. It connects relevant production assets implements all components that are necessary for the end-to-end digitalization of Industrie 4.0. Eclipse Basyx also supports, implements and constantly evaluates existing or still developing standards such as the administration shell of the Industrie 4.0 platform. (BaSyx Eclipse Wiki)

AAS: Asset Administration Shell

Asset Administration Shells (AAS) are one main concept of Industrie 4.0 for providing information hiding and higher levels of abstraction for assets. The AAS is the standardized self-description of a technical or logical component in production. Essentially, it is a machine-readable, technology & device-agnostic description of a component that provides access to all of its properties and functions.

AAS provides information about Industrie 4.0 assets, a unique identifier for this asset, and a generic interface to information regarding this Industrie 4.0 asset. They turn Assets into digitally manageable assets. (Adapted from BaSyx Eclipse Wiki)

MathWorks Simulink

Simulink is a MATLAB-based graphical programming environment for modeling, simulating and analyzing multi-domain dynamical systems. Its primary interface is a graphical block diagramming tool and a customizable set of block libraries. It offers tight integration with the rest of the MATLAB environment and can either drive MATLAB or be scripted from it. Simulink is widely used in automatic control and digital signal processing for multi-domain simulation and model-based design. (Mathworks)